Report for numerical breakdown of a forward recurrence

1. Introduction

In this assignment, the forward recurrence

$$I_n = nI_{n-1} - \frac{1}{e}$$

was implemented in Fortran and tested in single, double, and quadruple precision. The initial value was taken as

$$I_0 = 1 - e^{-1}.$$

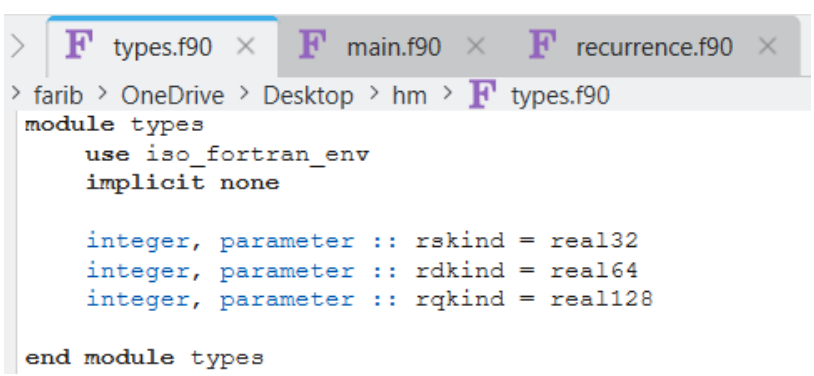
The aim was to determine when the recurrence starts to fail numerically in floating-point arithmetic. According to the theoretical properties of the integral, the sequence $I_n$should stay positive, strictly decreasing, and should approach zero as n increases. Therefore, the first numerical breakdown is detected when the computed value becomes non-positive or when the sequence stops decreasing.

2. Implementation

The program was written using three real kinds:

- real32 for single precision,
- real64 for double precision,
- real128 for quadruple precision.

Three separate subroutines were used to compute the recurrence in each precision. The main program tested the recurrence for three values of nmax: 20, 50, and 100.




```fortran
module types
    use iso_fortran_env
    implicit none

    integer, parameter :: rskind = real32
    integer, parameter :: rdkind = real64
    integer, parameter :: rqkind = real128

end module types
```

Fig1: Definition of floating-point kinds used in the program.

types.f90 × main.f90 × recurrence.f90 × test.f90

farib > OneDrive > Desktop > test.f90

```fortran
module recurrence
    use types
    implicit none

contains

subroutine single_precision(nmax)
    ...
end subroutine single_precision

subroutine double_precision(nmax)
    ...
end subroutine double_precision

subroutine quad_precision(nmax)
    ...
end subroutine quad_precision

end module recurrence
```

Fig2: Implementation steps of the forward recurrence in single, double, and quadruple precision.

types.f90 × main.f90 × recurrence.f9

farib > OneDrive > Desktop > hm > main.f90

```fortran
program numeric

    use types
    use recurrence
    implicit none

    integer :: i
    integer :: nmax_list(3)

    nmax_list = [20, 50, 100]

    print *, 'SINGLE PRECISION'

    do i = 1, 3
        print *, 'nmax = ', nmax_list(i)
        call single_precision(nmax_list(i))
    end do


    print *, 'DOUBLE PRECISION'

    do i = 1, 3
        print *, 'nmax = ', nmax_list(i)
        call double_precision(nmax_list(i))
    end do


    print *, 'QUAD PRECISION'

    do i = 1, 3
        print *, 'nmax = ', nmax_list(i)
        call quad_precision(nmax_list(i))
    end do

end program numeric
```

Fig3: Main program used to run the recurrence for different values of nmax.

3. Numerical Experiment

For each precision, the values of the recurrence were printed step by step. The code stopped when one of the following conditions occurred:

1. $I_n \leq 0$
2. $I_n \geq I_{n-1}$

From the actual program output, the first breakdown in all three cases occurred because the computed value became negative. This means that the first observed failure mode was loss of positivity, not loss of monotonicity.

```
fariba1184@DESKTOP-PQ1J1MT:/mnt/c/Users/farib/O
 SINGLE PRECISION
 nmax =           20
 step =            1  value =  0.264241099
 step =            2  value =  0.160602748
 step =            3  value =  0.113928795
 step =            4  value =   8.78357291E-02
 step =            5  value =   7.12991953E-02
 step =            6  value =   5.99157214E-02
 step =            7  value =   5.15305996E-02
 step =            8  value =   4.43653464E-02
 step =            9  value =   3.14086676E-02
 step =           10  value =  -5.37927747E-02
 algorithm failed at n =          10
 nmax =           50
 step =            1  value =  0.264241099
 step =            2  value =  0.160602748
 step =            3  value =  0.113928795
 step =            4  value =   8.78357291E-02
 step =            5  value =   7.12991953E-02
 step =            6  value =   5.99157214E-02
 step =            7  value =   5.15305996E-02
 step =            8  value =   4.43653464E-02
 step =            9  value =   3.14086676E-02
 step =           10  value =  -5.37927747E-02
 algorithm failed at n =          10
 nmax =          100
 step =            1  value =  0.264241099
 step =            2  value =  0.160602748
 step =            3  value =  0.113928795
 step =            4  value =   8.78357291E-02
 step =            5  value =   7.12991953E-02
 step =            6  value =   5.99157214E-02
 step =            7  value =   5.15305996E-02
 step =            8  value =   4.43653464E-02
 step =            9  value =   3.14086676E-02
 step =           10  value =  -5.37927747E-02
 algorithm failed at n =          10
```

Fig4: Output of the recurrence in single precision showing failure at $n = 10$.

```
DOUBLE PRECISION
nmax =            20
step =             1  value =  0.26424111765711533
step =             2  value =  0.16060279414278833
step =             3  value =  0.11392894125692266
step =             4  value =   8.7836323856248288E-002
step =             5  value =   7.1302178109799108E-002
step =             6  value =   5.9933627487352314E-002
step =             7  value =   5.1655951240023867E-002
step =             8  value =   4.5368168748748605E-002
step =             9  value =   4.0434077567295112E-002
step =            10  value =   3.6461334501508791E-002
step =            11  value =   3.3195238345154365E-002
step =            12  value =   3.0463418970410050E-002
step =            13  value =   2.8145005443888316E-002
step =            14  value =   2.6150635042994086E-002
step =            15  value =   2.4380084473468955E-002
step =            16  value =   2.2201910404060943E-002
step =            17  value =   9.5530356975936925E-003
step =            18  value = -0.19592479861475587
algorithm failed at n =           18
nmax =            50
step =             1  value =  0.26424111765711533
step =             2  value =  0.16060279414278833
step =             3  value =  0.11392894125692266
step =             4  value =   8.7836323856248288E-002
step =             5  value =   7.1302178109799108E-002
step =             6  value =   5.9933627487352314E-002
step =             7  value =   5.1655951240023867E-002
step =             8  value =   4.5368168748748605E-002
step =             9  value =   4.0434077567295112E-002
step =            10  value =   3.6461334501508791E-002
step =            11  value =   3.3195238345154365E-002
step =            12  value =   3.0463418970410050E-002
step =            13  value =   2.8145005443888316E-002
step =            14  value =   2.6150635042994086E-002
step =            15  value =   2.4380084473468955E-002
step =            16  value =   2.2201910404060943E-002
step =            17  value =   9.5530356975936925E-003
step =            18  value = -0.19592479861475587
algorithm failed at n =           18
```

```
QUAD PRECISION
nmax =            20
step =             1  value =  0.26424111765711535680895245967707825 5
step =             2  value =  0.16060279414278839202238114919269563 7
step =             3  value =  0.11392894125692285447161967741662603 7
step =             4  value =   8.7836323856249096290954939505043276 4E-0002
step =             5  value =   7.1302178109803159859250927363755509 2E-0002
step =             6  value =   5.9933627487376637559981794021072182 2E-0002
step =             7  value =   5.1655951240194141324348787986044403 0E-0002
step =             8  value =   4.5368168750110808999266533726894351 4E-0002
step =             9  value =   4.0434077579554959397875033380588290 2E-0002
step =            10  value =   3.6461334624107272383226563644422029 5E-0002
step =            11  value =   3.3195239693737674619968429927181452 2E-0002
step =            12  value =   3.0463435153409773844097388964716553 7E-0002
step =            13  value =   2.8145215822884738377742286379854325 3E-0002
step =            14  value =   2.6153580348944015692868239156499681 3E-0002
step =            15  value =   2.4424264062717913797499817186034346 5E-0002
step =            16  value =   2.2908783832044299164473304815088670 6E-0002
step =            17  value =   2.1569883973310764200522411695046528 2E-0002
step =            18  value =   2.0378470348151434013879640349376635 2E-0002
step =            19  value =   1.9311495443434924668189396476695196 8E-0002
step =            20  value =   1.8350467697256171768264159372443062 8E-0002
nmax =            50
step =             1  value =  0.26424111765711535680895245967707825 5
step =             2  value =  0.16060279414278839202238114919269563 7
step =             3  value =  0.11392894125692285447161967741662603 7
step =             4  value =   8.7836323856249096290954939505043276 4E-0002
step =             5  value =   7.1302178109803159859250927363755509 2E-0002
step =             6  value =   5.9933627487376637559981794021072182 2E-0002
step =             7  value =   5.1655951240194141324348787986044403 0E-0002
step =             8  value =   4.5368168750110808999266533726894351 4E-0002
step =             9  value =   4.0434077579554959397875033380588290 2E-0002
step =            10  value =   3.6461334624107272383226563644422029 5E-0002
step =            11  value =   3.3195239693737674619968429927181452 2E-0002
step =            12  value =   3.0463435153409773844097388964716553 7E-0002
step =            13  value =   2.8145215822884738377742286379854325 3E-0002
step =            14  value =   2.6153580348944015692868239156499681 3E-0002
step =            15  value =   2.4424264062717913797499817186034346 5E-0002
step =            16  value =   2.2908783832044299164473304815088670 6E-0002
step =            17  value =   2.1569883973310764200522411695046528 2E-0002
step =            18  value =   2.0378470348151434013879640349376635 2E-0002
step =            19  value =   1.9311495443434924668189396476695196 8E-0002
step =            20  value =   1.8350467697256171768264159372443062 8E-0002
step =            21  value =   1.7480380470937285538023576659843446 8E-0002
step =            22  value =   1.6688929189177960240994916355094957 4E-0002
step =            23  value =   1.5965930179650763947359306005723147 2E-0002
step =            24  value =   1.5302883140176013141099573975894660 8E-0002
step =            25  value =   1.4692637332958006931965579235905646 2E-0002
step =            26  value =   1.4129129485465858635581289972085929 7E-0002
step =            27  value =   1.3607054936135861565171059084859229 5E-0002
```

```
nmax =           100
step =             1  value =  0.26424111765711533
step =             2  value =  0.16060279414278833
step =             3  value =  0.11392894125692266
step =             4  value =   8.7836323856248288E-002
step =             5  value =   7.1302178109799108E-002
step =             6  value =   5.9933627487352314E-002
step =             7  value =   5.1655951240023867E-002
step =             8  value =   4.5368168748748605E-002
step =             9  value =   4.0434077567295112E-002
step =            10  value =   3.6461334501508791E-002
step =            11  value =   3.3195238345154365E-002
step =            12  value =   3.0463418970410050E-002
step =            13  value =   2.8145005443888316E-002
step =            14  value =   2.6150635042994086E-002
step =            15  value =   2.4380084473468955E-002
step =            16  value =   2.2201910404060943E-002
step =            17  value =   9.5530356975936925E-003
step =            18  value = -0.19592479861475587
algorithm failed at n =           18
```

```
step =            28  value =   1.3118097040361802229265884214597553 2E-0002
step =            29  value =   1.2545372999049943053186872061868171 0E-0002
step =            30  value =   8.4817488000559700000823916945842561 0E-0003
step =            31  value = -0.10494522836970725159296962762934833 3
algorithm failed at n =           31
nmax =           100
step =             1  value =  0.26424111765711535680895245967707825 5
step =             2  value =  0.16060279414278839202238114919269563 7
step =             3  value =  0.11392894125692285447161967741662603 7
step =             4  value =   8.7836323856249096290954939505043276 4E-0002
step =             5  value =   7.1302178109803159859250927363755509 2E-0002
step =             6  value =   5.9933627487376637559981794021072182 2E-0002
step =             7  value =   5.1655951240194141324348787986044403 0E-0002
step =             8  value =   4.5368168750110808999266533726894351 4E-0002
step =             9  value =   4.0434077579554959397875033380588290 2E-0002
step =            10  value =   3.6461334624107272383226563644422029 5E-0002
step =            11  value =   3.3195239693737674619968429927181452 2E-0002
step =            12  value =   3.0463435153409773844097388964716553 7E-0002
step =            13  value =   2.8145215822884738377742286379854325 3E-0002
step =            14  value =   2.6153580348944015692868239156499681 3E-0002
step =            15  value =   2.4424264062717913797499817186034346 5E-0002
step =            16  value =   2.2908783832044299164473304815088670 6E-0002
step =            17  value =   2.1569883973310764200522411695046528 2E-0002
step =            18  value =   2.0378470348151434013879640349376635 2E-0002
step =            19  value =   1.9311495443434924668189396476695196 8E-0002
step =            20  value =   1.8350467697256171768264159372443062 8E-0002
step =            21  value =   1.7480380470937285538023576659843446 8E-0002
step =            22  value =   1.6688929189177960240994916355094957 4E-0002
step =            23  value =   1.5965930179650763947359306005723147 2E-0002
step =            24  value =   1.5302883140176013141099573975894660 8E-0002
step =            25  value =   1.4692637332958006931965579235905646 2E-0002
step =            26  value =   1.4129129485465858635581289972085929 7E-0002
step =            27  value =   1.3607054936135861565171059084859229 5E-0002
step =            28  value =   1.3118097040361802229265884214597553 2E-0002
step =            29  value =   1.2545372999049943053186872061868171 0E-0002
step =            30  value =   8.4817488000559700000823916945842561 0E-0003
step =            31  value = -0.10494522836970725159296962762934833 3
algorithm failed at n =           31
```

Fig5: Output of the recurrence in double precision showing failure at $n = 18$ and recurrence in quadruple precision showing failure at $n = 31$.

4. Results

The results are summarized as below.

Precision	Tested nmax values	Observed n_{crash}	Failure type
Single precision	20, 50, 100	10	Loss of positivity
Double precision	20, 50, 100	18	Loss of positivity
Quadruple precision	20, 50, 100	31	Loss of positivity

5. Interpretation

In all three precisions, the recurrence behaves reasonably well at the beginning: the values are positive and decrease gradually. However, after a certain number of steps, the computed solution suddenly becomes negative. Since the exact integral is positive for all n, this is an unphysical result and indicates numerical breakdown.

The difference between the three precisions is also clear. In single precision, the breakdown happens earliest at $n = 10$. In double precision, it is delayed until $n = 18$. In quadruple precision, the recurrence remains stable for much longer and fails only at $n = 31$. This trend is expected, because higher precision reduces the size of the initial rounding error. The main reason for this behaviour is the propagation of numerical error through the recurrence. If the initial value contains a small error ε_0, then after n steps the propagated error satisfies

$$\varepsilon_n = n!\,\varepsilon_0.$$

This means that the error grows factorially, which is extremely fast. Even if the starting error is very small, repeated application of the recurrence multiplies it so strongly that after some point it becomes comparable to the exact value of the integral. At that stage, the numerical solution can no longer preserve the true behaviour of the sequence and the computed values become unreliable.

Another useful observation from the output is that the true values are getting smaller as n increases, especially before the crash point. This makes the recurrence even more sensitive, because once the exact solution becomes small, it takes only a moderate numerical error to completely change the sign of the result. That is why the failure appears as a sudden jump to a negative value.

These results show that increasing the precision does help, but only temporarily. Moving from single to double and then to quadruple precision delays the breakdown from 10 to 18 and then to 31 steps. However, the instability is not removed. This confirms the main idea of the exercise: the problem is not a coding mistake, but the numerical instability of the forward recurrence itself.